



Brain Tumor Classification with Vision Transformers

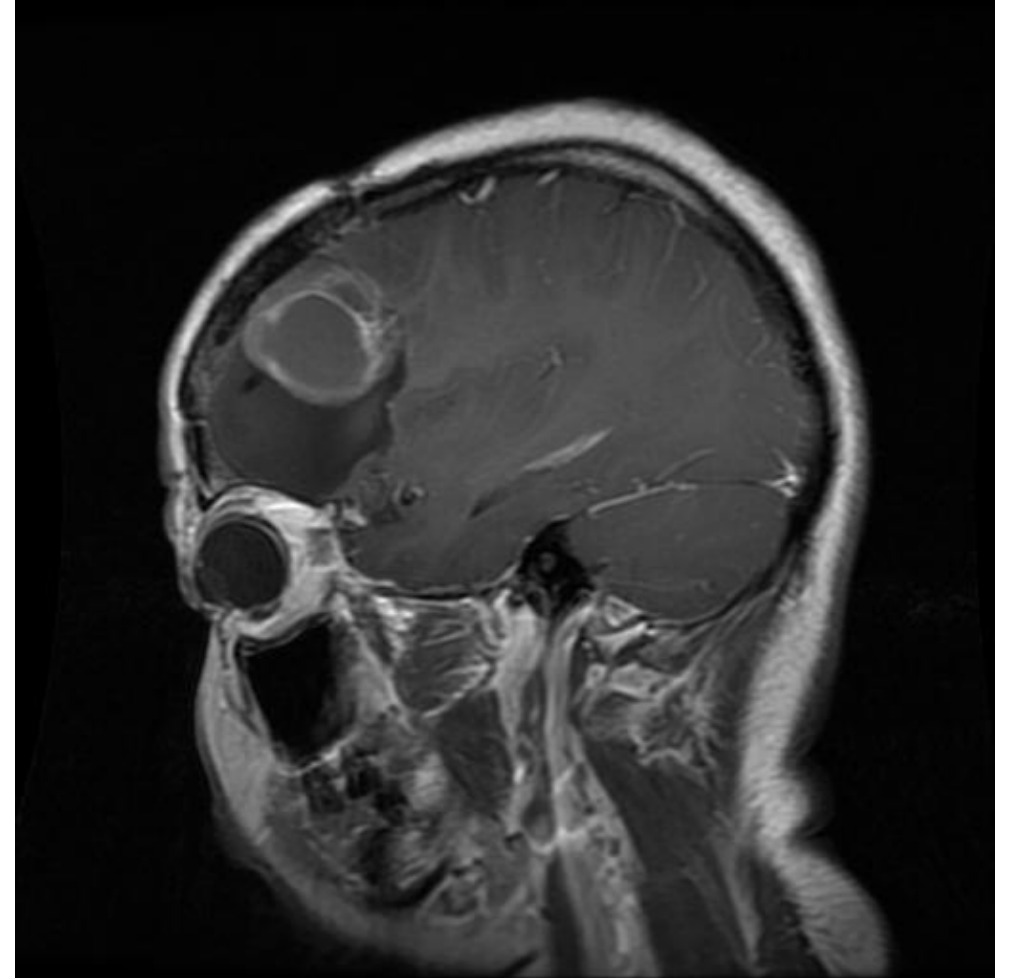
Investigating the performance of polar coordinate positional embeddings (PoPE)

AML2026 Group 09:

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Brain Tumor Classification

- most serious neurological disease
- Early diagnosis is critical
- Magnetic Resonance Imaging is the state of Art
- Manual MRI analysis is challenging



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Brain Tumor Classification and Deep learning

- CNNs used traditionally
- CNNs capture local image features
- Transformers might capture long-range spatial relationships

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Brain Tumor Classification and Deep learning

- CNNs used traditionally
- CNNs capture local image features
- Transformers might capture long-range spatial relationships
- Transformer-based glioma MRI analysis shows:
 - Large receptive field
 - Capture relationships between tumor and surrounding tissue
 - Global contextual information learning through self-attention mechanisms

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Our Project

- Multi-class brain tumor MRI classification
 - For a given MRI image, the goal is to predict a label $y \in \{\text{glioma, meningioma, pituitary, no tumor}\}$
- Vision transformer encoder with PoPe (Polar Coordinate Positional Embeddings)
 - PoPe might help disentangle
 - Positional information (where is the tumor)
 - Content information (what does the tumor look like)
- Encoder outputs are passed to an MLP classifier

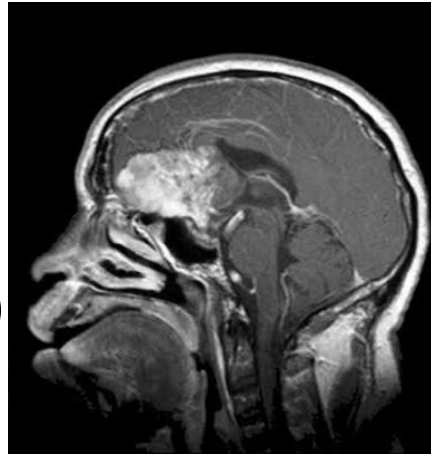
Evaluation Protocol:

- Vision transformer encoder with RoPe used as a baseline
- AUROC as main metric

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Our Project

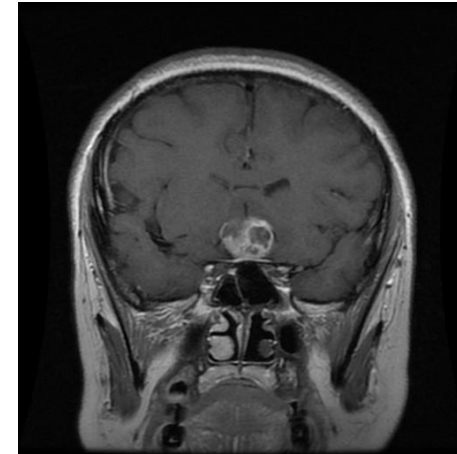
- Publicly available brain MRI dataset from Kaggle
 - contains multiple clinically relevant tumor categories
 - heterogenous scans in terms of shape, location and growth pattern of tumors
- 4 categories of scans:
 - Glioma (N = 926)
 - Meningioma (N = 937)
 - Pituitary Tumor (N = 932)
 - No Tumor (N = 469)



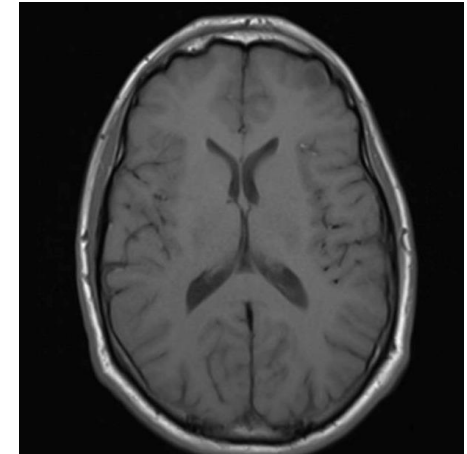
Glioma



Meningioma



Pituitary Tumor



No Tumor

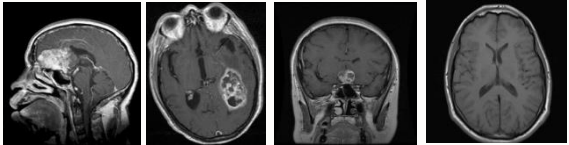
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Data Pipeline & Augmentation

RAW POOL

Brain MRI — labelled images

Original Training + Testing folders merged, then re-split.



STRATIFIED SPLIT · PER CLASS SHUFFLE

Train · Val · Test

train 80%

20%

20%

Label distribution held identical across all three splits. Test split = 325scans.

CLASS IMBALANCE HANDLING

Inverse-frequency class weights

Normalised so mean weight = 1, fed into weighted cross-entropy.

TRAIN TRANSFORMS · STOCHASTIC

Resize 224 × 224
bilinear, antialias

EnsureRGB
1-ch → 3-ch · drop alpha

RandomAffine AUG
±15° · translate 5% · scale .95-1.05 · shear 5°

RandomHorizontalFlip AUG
p = 0.5

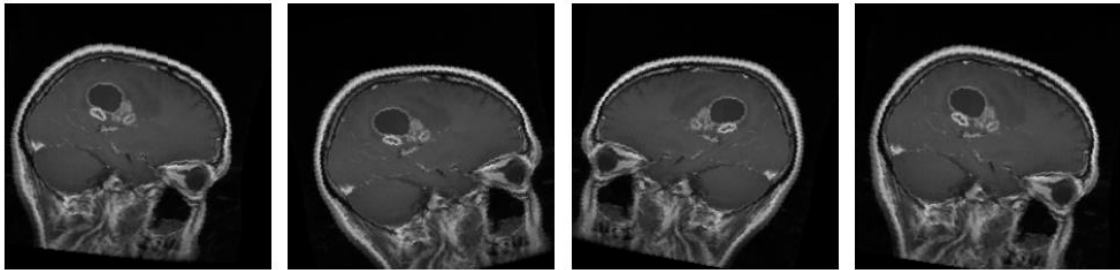
ToDtype float32
scale to [0,1]

Normalize
per-split μ , σ

VARIANTS_PER_IMAGE × 3



Augmented variants — glioma_tumor



VAL / TEST TRANSFORMS · DETERMINISTIC

Resize 224 × 224
bilinear, antialias

EnsureRGB
1-ch → 3-ch

ToDtype float32
scale to [0,1]

Normalize
per-split μ , σ

DATALOADERS

Batched tensors → models

batch 32

train: shuffle

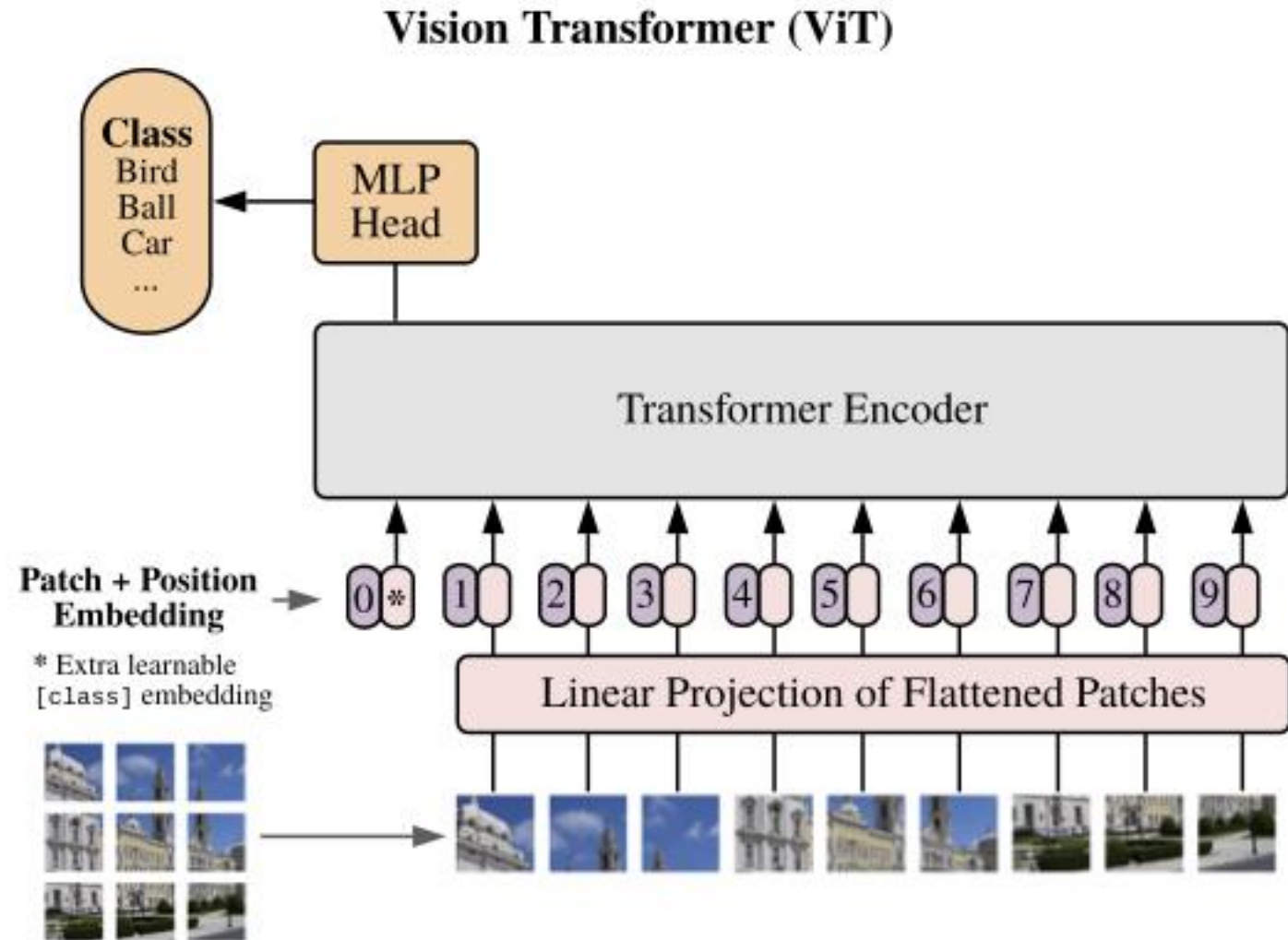
val/test: ordered

Output (B, 3, 224, 224) float32, normalized.

Vision Transformer

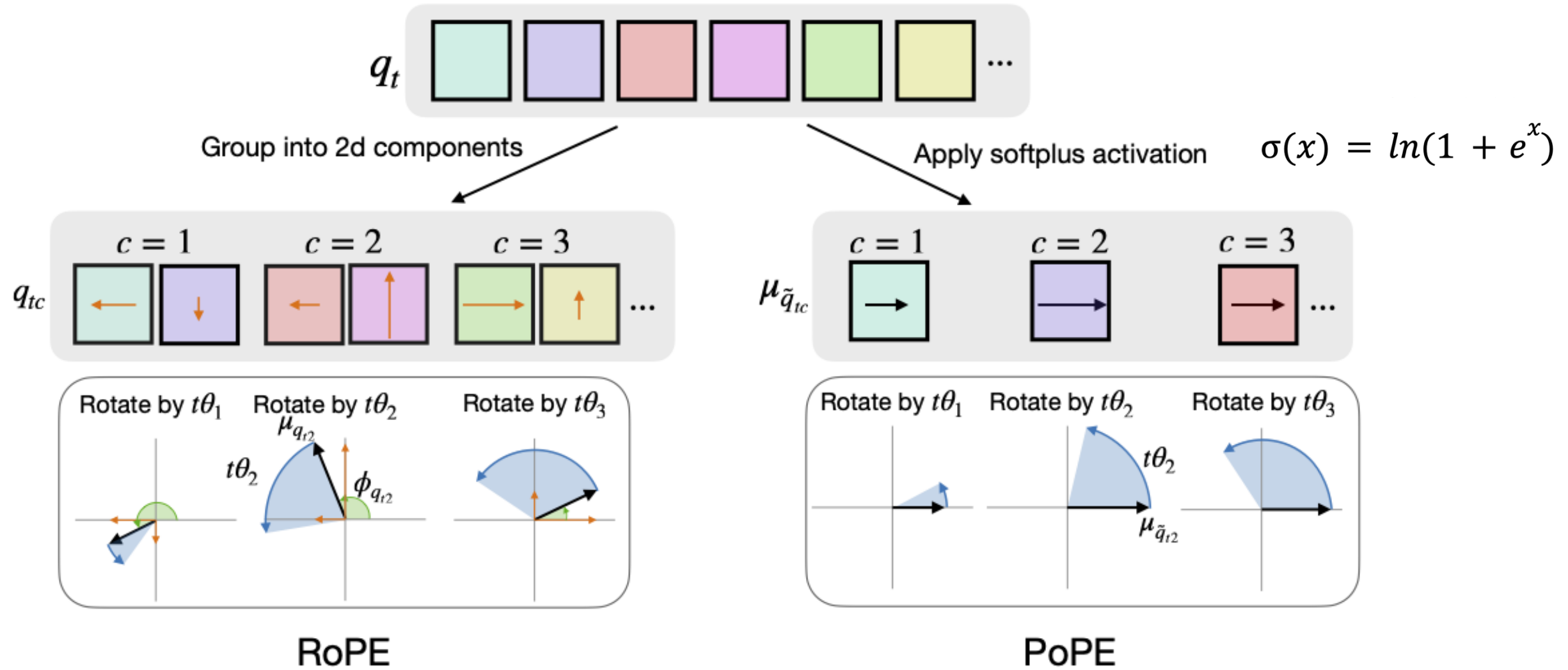
A Vision Transformer can be used for image classification by:

- reshaping the image into a sequence of flattened 2D Patches
- adding positional embeddings
- feeding the embedding of special token [class] to an MLP



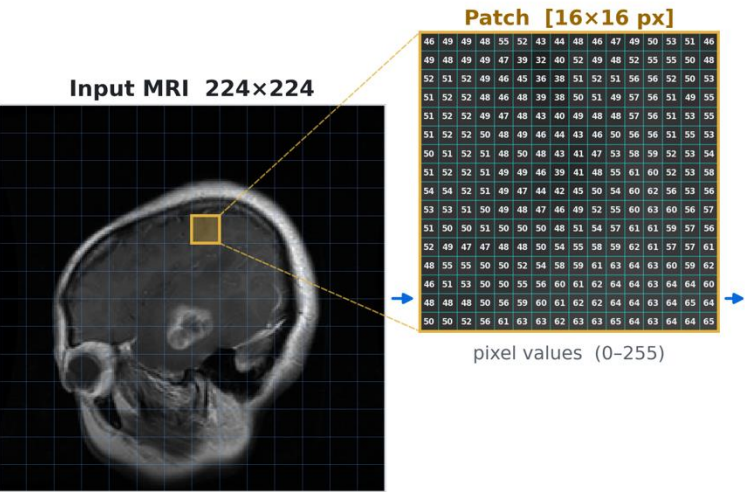
Source: Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv. <https://doi.org/10.48550/ARXIV.2010.11929>

The Intuition behind PoPE



Source: Gopalakrishnan, A., Csordás, R., Schmidhuber, J., & Mozer, M. C. (2025). Decoupling the ‘what’ and ‘where’ with polar coordinate positional embeddings. arXiv (Preprint). <https://doi.org/10.48550/ARXIV.2509.10534>

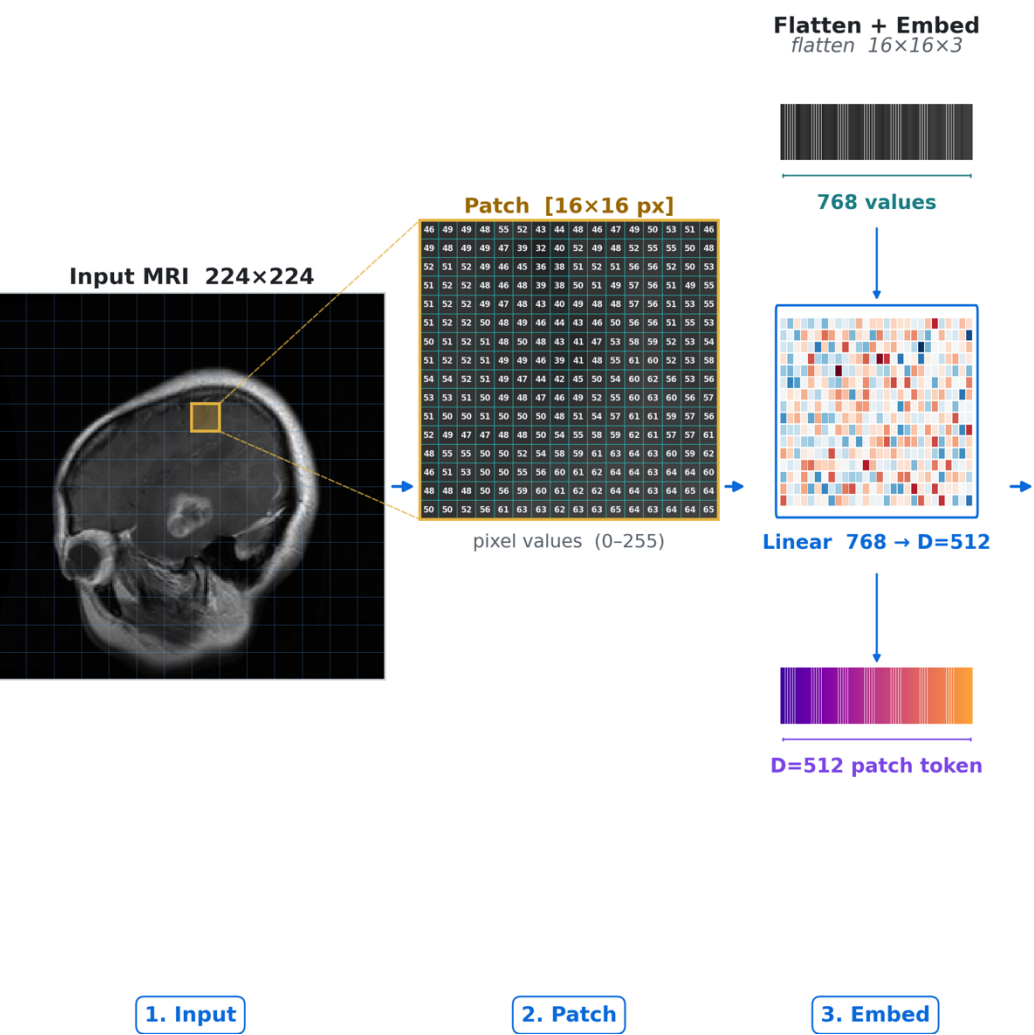
Forward Pass of ViT – Glioma Tumor Example



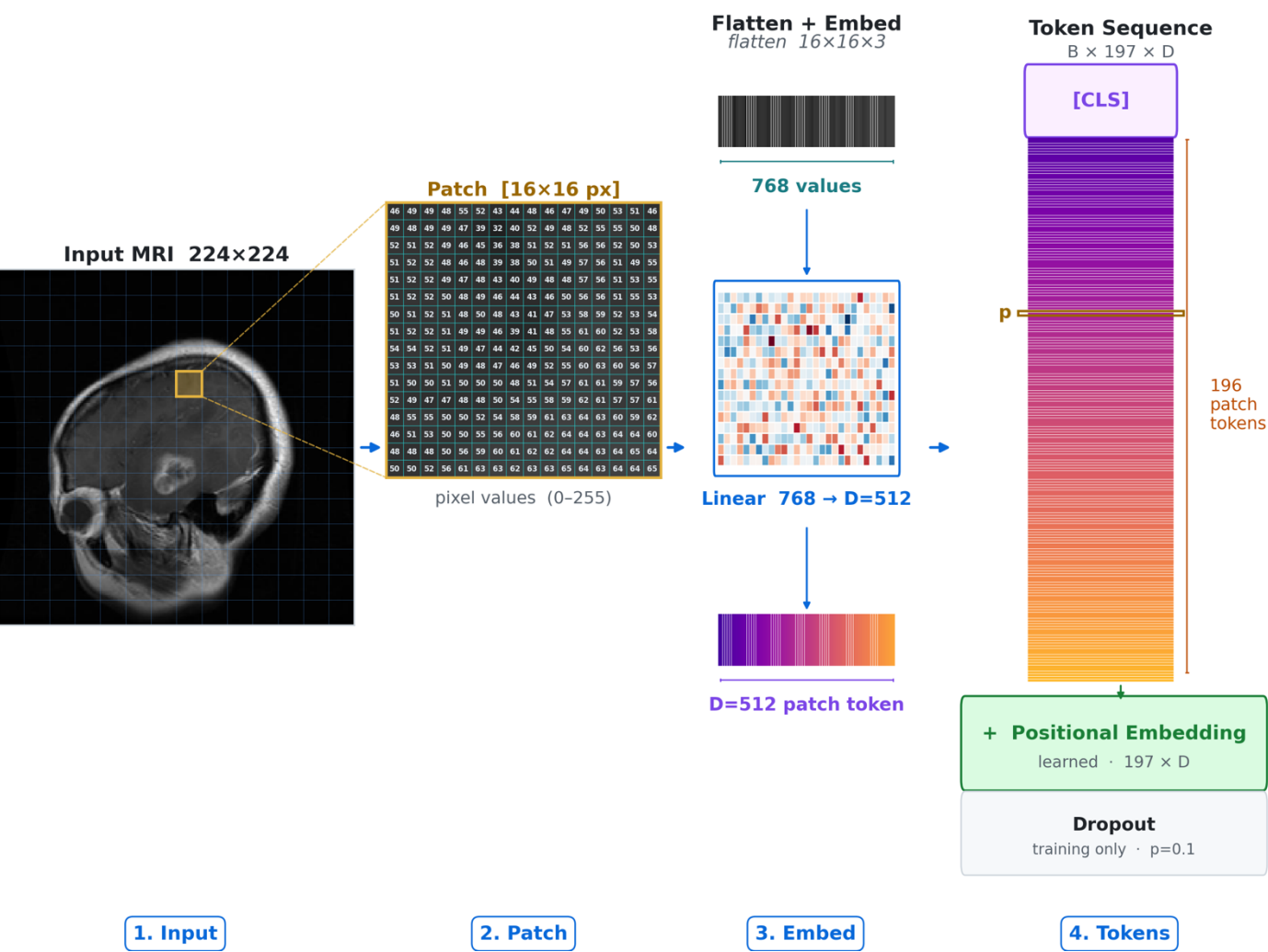
1. Input

2. Patch

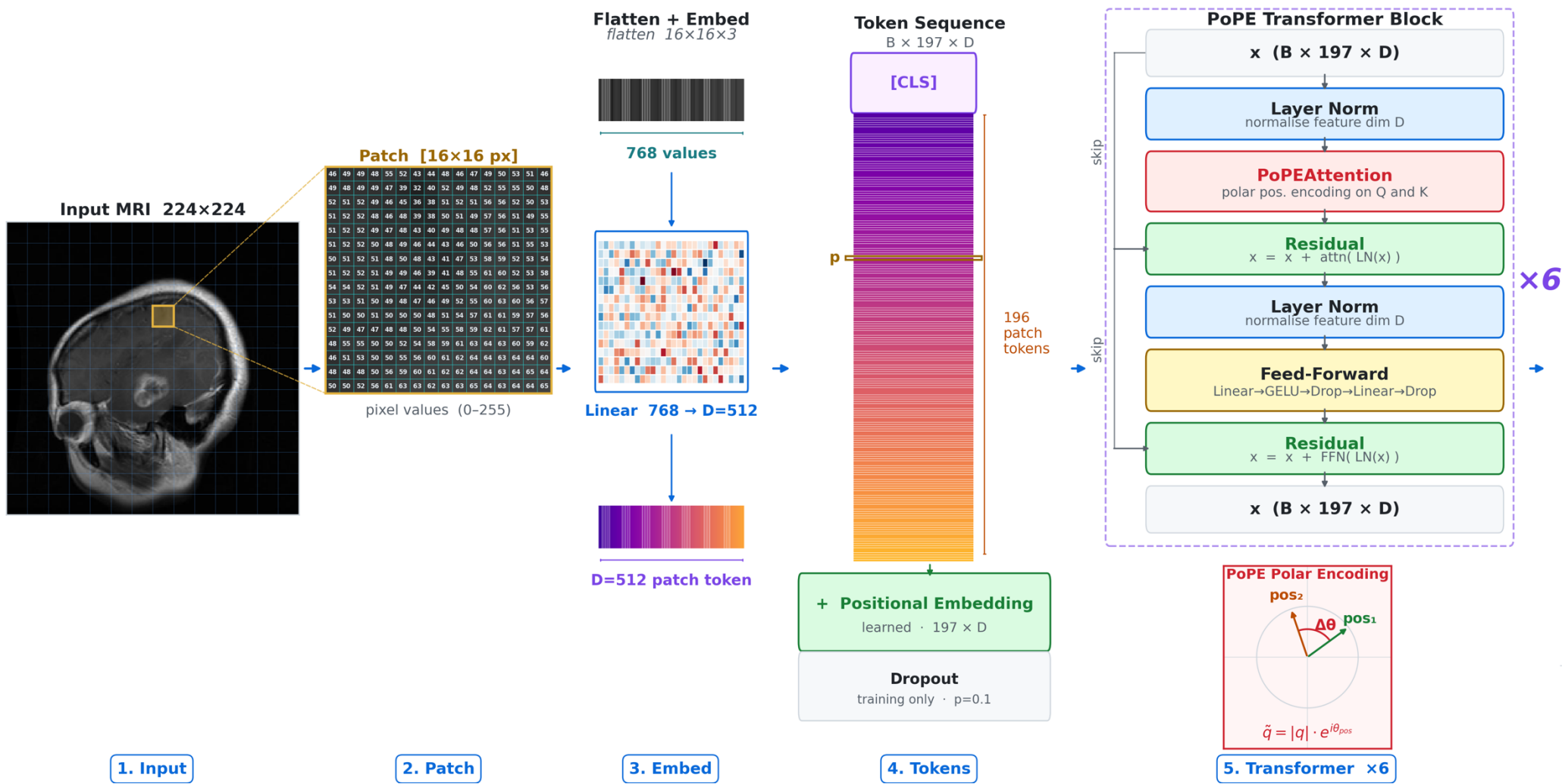
Forward Pass of ViT – Glioma Tumor Example



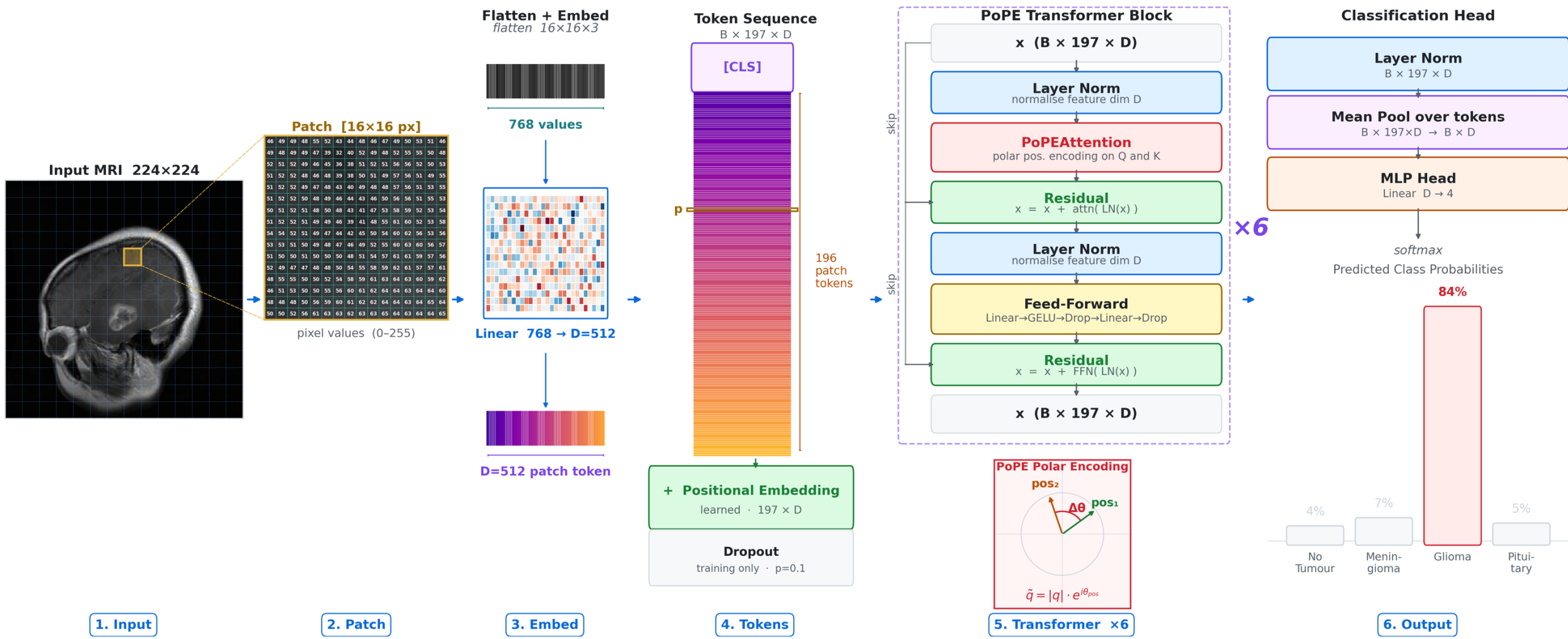
Forward Pass of ViT – Glioma Tumor Example



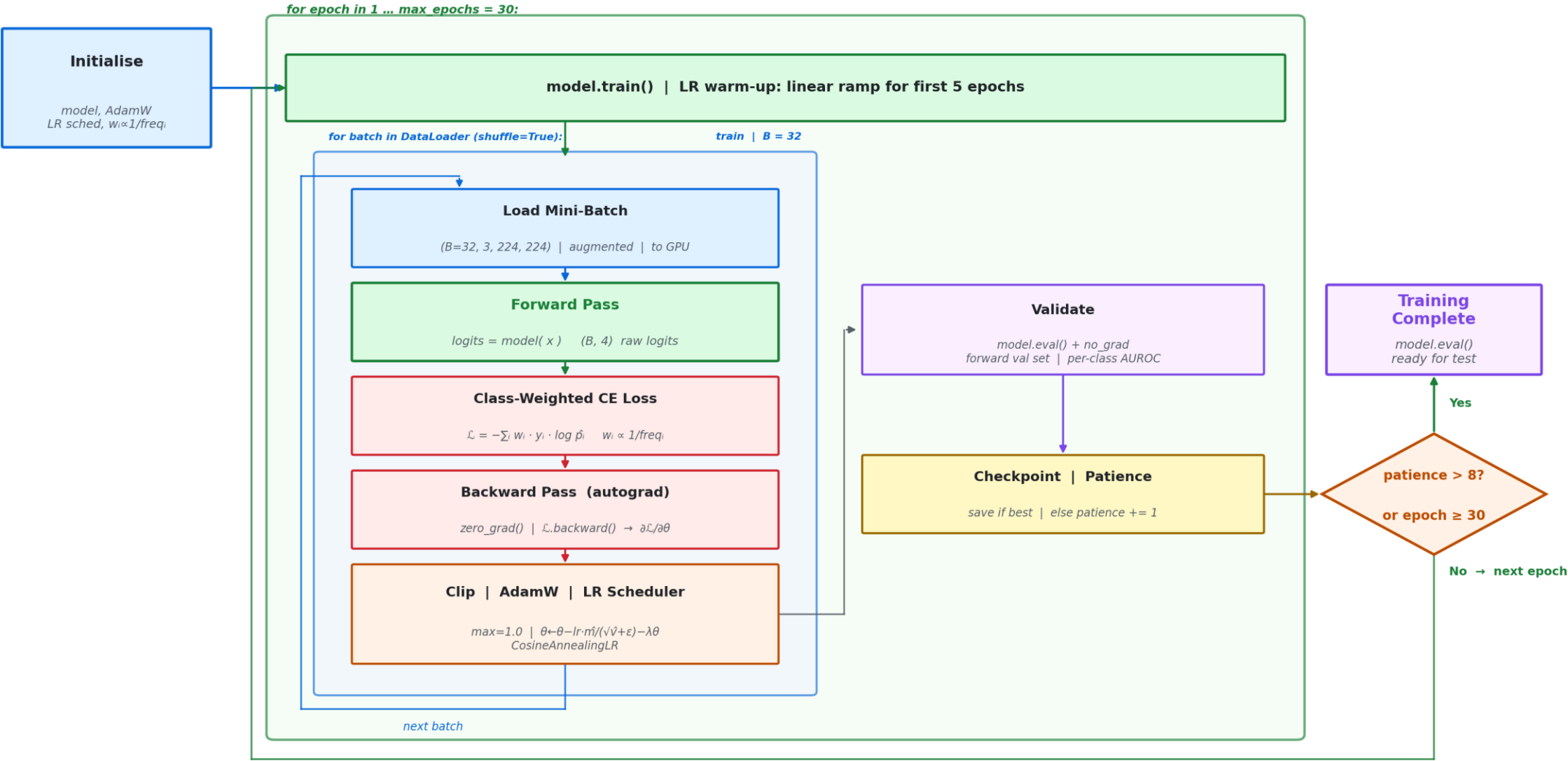
Forward Pass of ViT – Glioma Tumor Example



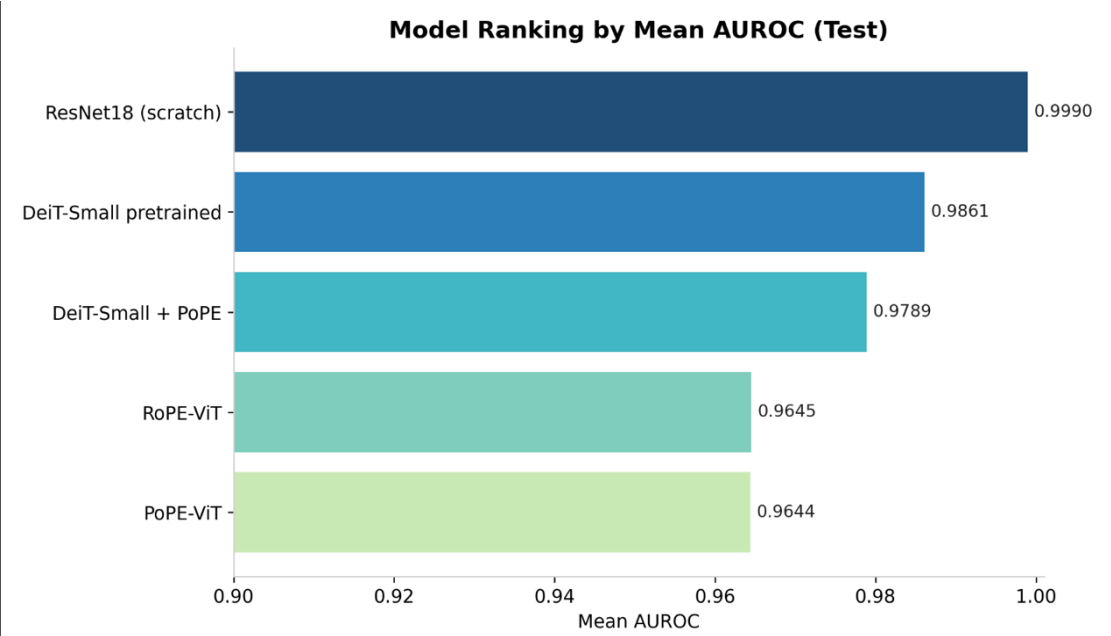
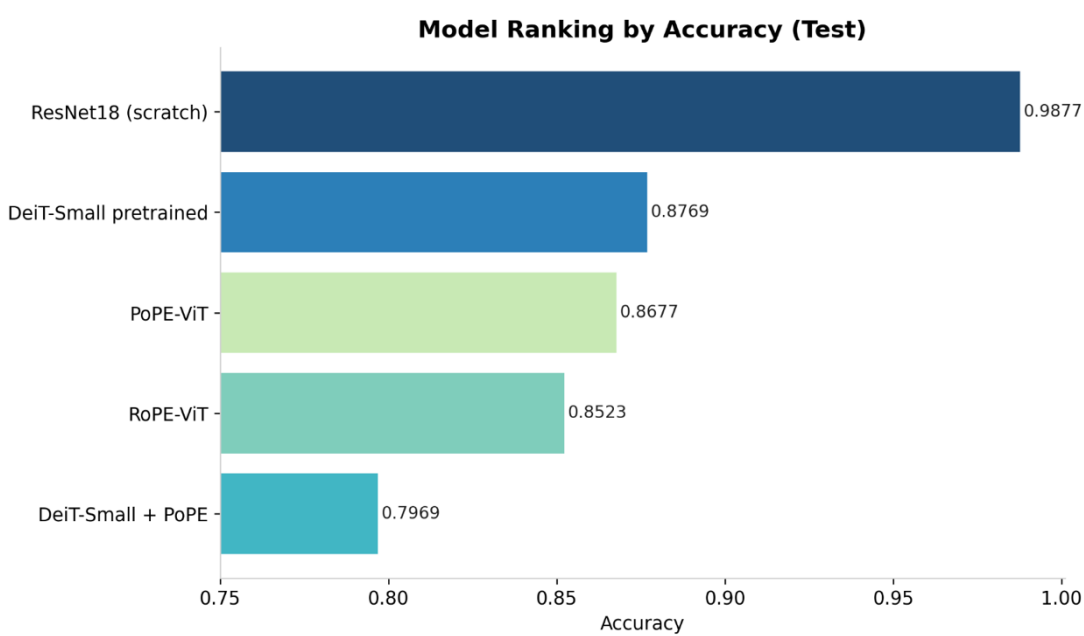
Forward Pass of ViT – Glioma Tumor Example



Training Pipeline

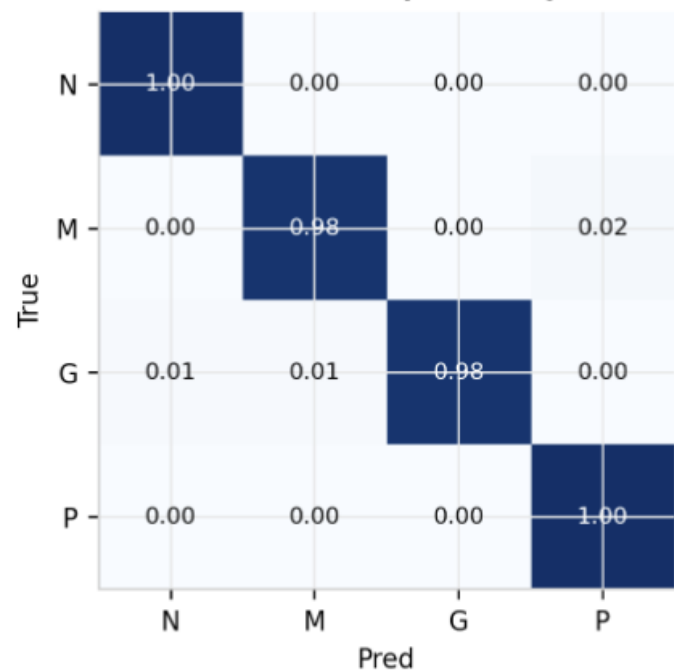


Results

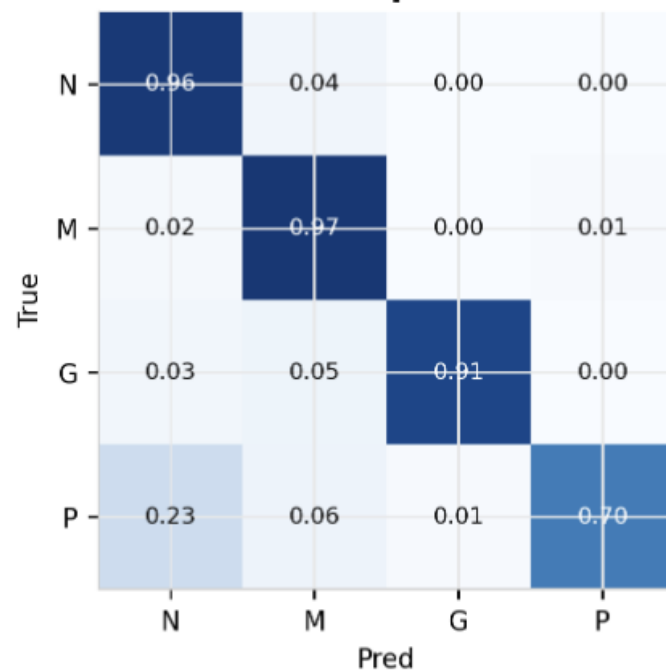


Confusion Matrices (Test, row-normalized)

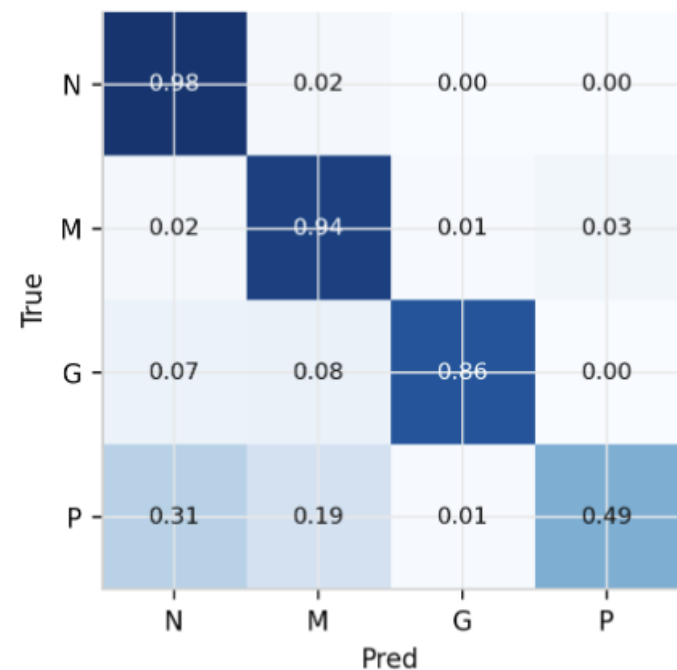
ResNet18 (scratch)



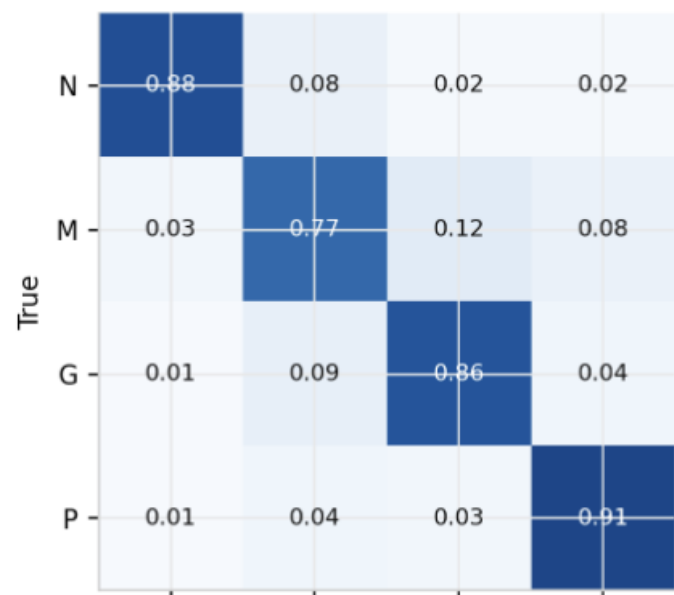
DeiT-Small pretrained



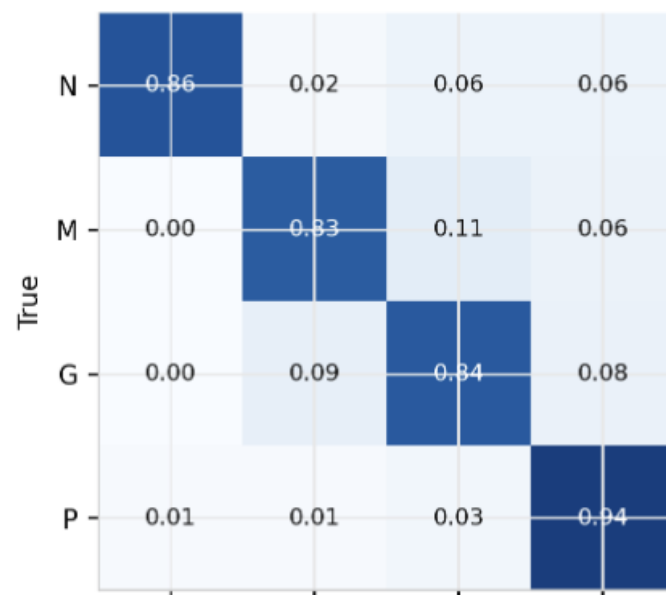
DeiT-Small + PoPE



RoPE-ViT



PoPE-ViT

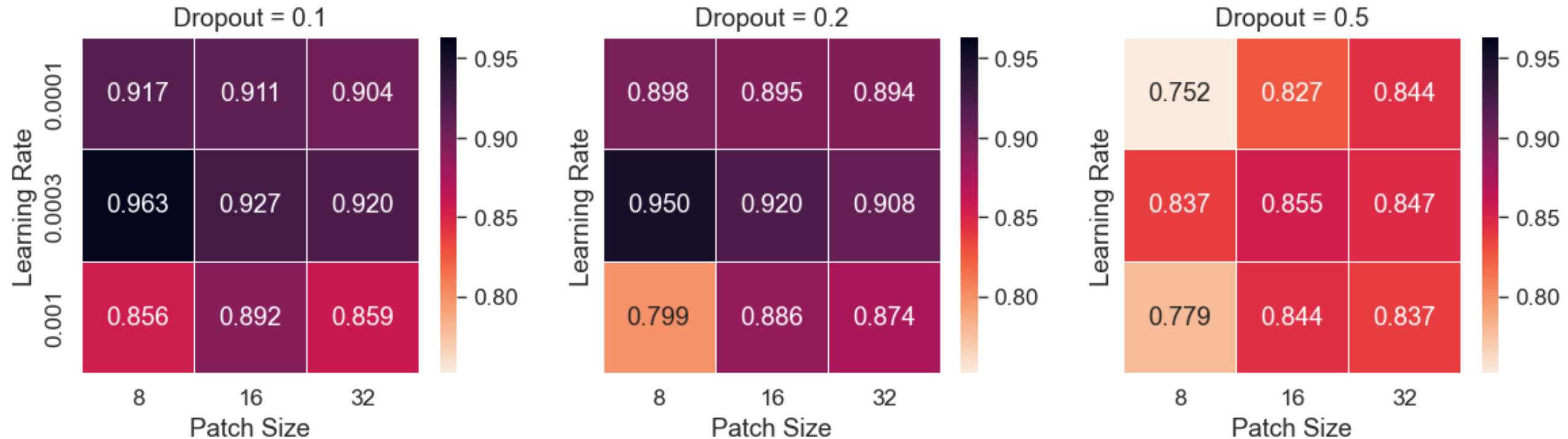


Results: Hyperparameter Tuning

Grid search over patch size, dropout and learning rate:

- Best setting: Patch size: 8, Dropout: 0.1, Learning rate: $3e-4$
- Performance decreases with increasing dropout percentage
- Medium learning rate or $3e-4$ was ideal
- Patch size has different effects at different learning rates

Grid Search AUROC Results



Limitations

- A key limitation is that our core custom ViT experiments (RoPE-ViT and PoPE-ViT) were trained from scratch rather than initialized from a large-scale pretrained visual transformer backbone, which may have limited their final performance and generalization.
- One limitation of the dataset is that multiple MRI slices originate from the same patient. Since the dataset does not provide patient-level identifiers, the data split was performed at the image level. Consequently, images from the same patient may appear in both training and testing sets, which could lead to optimistic performance estimates.

The Intuition behind PoPE: Mathematical Derivation

Core observation of Gopalakrishnan et al. 2025:

In RoPE, for a query vector q at time step t and a key vector k at time step s , every component q_{tc} and k_{sc} at component index c can be expressed as a complex number in polar form:

$$q_{tc} \Leftrightarrow (\mu_{q_{tc}}, \phi_{q_{tc}})$$

$$k_{sc} \Leftrightarrow (\mu_{k_{sc}}, \phi_{k_{sc}})$$

The attention score between the RoPE-embedded vectors is then computed as the real part of the complex inner product between $RoPE(q_{tc})$ and $RoPE(k_{sc})$:

$$a_{ts}^{RoPE} = Re(\overline{RoPE(q_{tc})} RoPE(k_{sc}))$$

Which simplifies to

$$a_{ts}^{RoPE} = \mu_{q_{tc}} \mu_{k_{sc}} \cos((s - t)\theta_c + \phi_{k_{sc}} - \phi_{q_{tc}})$$

The first term $(\mu_{q_{tc}} \mu_{k_{sc}})$ represents the similarity between the content of the two vectors. Ideally we would want the second term to capture only the similarity between positions (i.e. the relative position $(s - t)$), but it also depends on the initial angle $\phi_{q_{tc}}$ and $\phi_{k_{sc}}$, which is a function of the content of the vectors.

In PoPE, the initial initial angles are set to 0, so that the attention score simplifies to:

$$a_{ts}^{PoPE} = \mu_{q_{tc}} \mu_{k_{sc}} \cos((s - t)\theta_c)$$

Source: Gopalakrishnan, A., Csordás, R., Schmidhuber, J., & Mozer, M. C. (2025). Decoupling the ‘what’ and ‘where’ with polar coordinate positional embeddings. arXiv (Preprint). <https://doi.org/10.48550/ARXIV.2509.10534>